

Azure vs AWS Cost Comparison Report

1. Introduction

Cloud computing has become an essential component of modern application deployment, offering scalability, flexibility, and cost efficiency. This project evaluates the estimated cost of hosting a small web application using two leading cloud platforms: Amazon Web Services (AWS) and Microsoft Azure.

The objective of this analysis is to compare equivalent services across both providers, estimate monthly and annual costs using their respective pricing calculators, evaluate available discount mechanisms, and identify cost optimization opportunities. The findings help determine which cloud provider offers the most cost-effective solution for a small business or startup environment.

2. Application Requirements

The web application used for this comparison was configured with the following requirements:

Requirement	Specification
Virtual Machine	1 Instance
CPU	2 vCPUs
Memory	4 GB RAM
Storage	100 GB SSD
Database	Managed Relational Database
Data Transfer	200 GB Outbound Traffic
Operating System	Linux
Billing Model	Pay-As-You-Go

Service Equivalency Mapping

Requirement	AWS Service	Azure Service
Virtual Machine	Amazon EC2	Azure Virtual Machines
Storage	Amazon Elastic Block Store (EBS)	Azure Managed Disk
Database	Amazon RDS	Azure SQL Database
Data Transfer	AWS Data Transfer	Azure Bandwidth

This mapping ensures that equivalent services are compared across both platforms.

3. AWS Cost Estimate

The AWS estimate was generated using the AWS Pricing Calculator with the following services:

Service	Monthly Cost (USD)
Amazon EC2	\$15.04
Amazon EBS	\$19.49
Amazon RDS	\$778.11
AWS Data Transfer	\$2.00
Total Monthly Cost	\$814.64

Annual Cost

Cost Type	Amount
Monthly Cost	\$814.64
Annual Cost	\$9,775.68

4. Azure Cost Estimate

The Azure estimate was generated using the Microsoft Azure Pricing Calculator with equivalent services.

Service	Monthly Cost (USD)
Azure Virtual Machine	\$77.02
Azure SQL Database	\$375.66
Managed Disk and Data Transfer	Included
Total Monthly Cost	\$452.68

Annual Cost

Cost Type	Amount
Monthly Cost	\$452.68
Annual Cost	\$5,432.16

5. Cost Comparison Table

Monthly Cost Comparison

Category	AWS (USD)	Azure (USD)
Compute	\$15.04	\$77.02
Storage	\$19.49	Included
Database	\$778.11	\$375.66
Data Transfer	\$2.00	Included
Total Monthly Cost	\$814.64	\$452.68

Annual Cost Comparison

Provider	Annual Cost (USD)
AWS	\$9,775.68
Azure	\$5,432.16

Cost Difference Analysis

Metric	Value
Monthly Difference	\$361.96
Annual Difference	\$4,343.52
Percentage Savings with Azure	44.43%

Analysis

The results show that Microsoft Azure provides a lower overall monthly and annual cost for the selected workload. Although Azure's virtual machine cost is higher than AWS EC2, the Azure SQL Database service is significantly less expensive than the Amazon RDS configuration used in this comparison. As a result, Azure offers a total monthly saving of approximately 44.43%.

6. Discount Mechanisms Comparison

Cloud providers offer various discount programs to reduce long-term operational costs.

AWS	Azure
AWS Savings Plans	Azure Reserved VM Instances

AWS	Azure
Reserved Instances	Azure Reserved Capacity
AWS Free Tier	Azure Free Account
Volume Discounts	Enterprise Agreements
Long-Term Commitments	Azure Hybrid Benefit

Analysis

AWS Savings Plans and Reserved Instances can significantly reduce compute costs for predictable workloads. Azure offers similar benefits through Reserved VM Instances and Azure Hybrid Benefit, which allows organizations with existing Microsoft licenses to reduce licensing costs.

Both providers encourage long-term commitments by offering discounted pricing compared to standard pay-as-you-go rates.

7. Cost Optimization Strategies

Strategy 1: Right-Sizing Resources

Organizations should continuously monitor resource utilization and select virtual machine sizes that match actual workload requirements. Over-provisioned resources increase costs without providing additional value.

Strategy 2: Use Reserved Capacity

Businesses with predictable workloads can lower costs by purchasing Reserved Instances, Savings Plans, or Reserved VM Capacity instead of relying solely on pay-as-you-go pricing.

Strategy 3: Monitor and Remove Unused Resources

Regular audits should be conducted to identify unused virtual machines, storage volumes, databases, and network resources. Removing unnecessary services reduces monthly expenditure.

Strategy 4: Optimize Database Resources

Database services represent the largest portion of the total cost in this comparison. Selecting appropriate database tiers and storage configurations can result in significant savings.

8. Business Context Summary

For startups and small businesses, cloud infrastructure costs can have a substantial impact on operational budgets. Choosing a cost-effective cloud provider allows organizations to allocate more resources toward product development, marketing, and customer acquisition.

Based on the estimates generated in this project, Microsoft Azure offers a lower overall cost than AWS for the specified workload. Azure's monthly cost of \$452.68 is considerably lower than AWS's monthly cost of \$814.64, resulting in annual savings of approximately \$4,343.52.

Organizations already invested in Microsoft technologies may benefit further from Azure's integration with Windows Server, Active Directory, Microsoft 365, and Azure Hybrid Benefit. Conversely, AWS offers a mature ecosystem with extensive services and global availability.

While Azure appears to be the more economical option for this workload, the final provider selection should also consider factors such as performance requirements, organizational expertise, scalability needs, compliance requirements, and service availability.

9. Conclusion

This project compared the estimated cost of hosting a small web application on AWS and Microsoft Azure using equivalent cloud services. Cost estimates were generated using the official AWS Pricing Calculator and Azure Pricing Calculator.

The analysis showed that Azure provides a lower total monthly cost of **\$452.68** compared to AWS's **\$814.64**. This represents a potential cost saving of approximately **44.43%** and an annual saving of **\$4,343.52**.

Both platforms offer robust cloud services, flexible pricing models, and discount programs for long-term workloads. However, based on the cost estimates obtained for this project, **Microsoft Azure is the more cost-effective option for the selected application requirements.**